

Invited Lecture

Multiprocessor Revolution and Cache Management

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Abstract

Advancement in semiconductor technology is allowing to pack more and more transistors on a single die, thus increasing the complexity of the systems. Chip density is continuing to increase approximately twice every two years. However, there is little instruction level parallelism left to improve the performance of a single processor. The clock speeds are also not increasing due to heat dissipation issues. The solution is to increase the number of cores on a chip instead of improving performance of single cores. More number of simple and small cores running at relatively slower clock rate can help to improve system performance by exploiting thread level parallelism. Threads of one or more applications can run on these multiple cores and communicate with each other to provide us with the end result. Communication among threads can be established using either message passing or shared memory.

Chip multiprocessors have 2 to 3 levels of cache hierarchy and hence management of shared data among different caches becomes crucial for correctness and performance. Caches are mainly characterised as *private* or *shared*. Private caches provide faster access but limit the amount of data that can be stored. Shared caches on the other hand provide larger (cumulative) capacity but suffer from non-uniform access latencies. Thus, the main aspects that govern cache management are the *access latency* and cache *space utilization*. In this talk we try to address these issues by forming logical clusters of caches (and cores); and by increasing the lifetime of a cache blocks on the chip. This leads to multiple clusters on the chip, and in turn multiple copies of cache blocks among these clusters. We therefore need a new method of managing coherence of data among the clusters.

The talk will discuss the multiprocessor revolution; issues related to shared cache accesses; and our proposed technique for cache management and the modified cache coherence protocol.